Grade 10 – Term 1: C8 Classroom Activity

Guided Problems

1. Cold-storage cooperative. A produce cooperative chooses a storage system for the harvest season.

Payoffs

Alt.	Strong	Normal	Weak
p	0.4	0.4	0.2
A_1	96	48	-12
A_2	72	60	18
A_3	48	42	34

Summary

Alt.	Max	Min	Regret	EMV
A_1	96	-12	46	55.2
A_2	72	18	24	56.4
A_3	48	34	48	42.8

Decisions

Criterion	A_i	Value
Maximax		
Maximin		
Regret		
EMV		

A_1: New warehouse A_2: Modular storage A_3: Rented storage

Task 1. Identify the four original recommendations and deciding values; include ties or mark EMV/Bayes unavailable.
Task 2. A grant changes A_2's weak-demand payoff to $18 + s$, where s is an integer, $0 \le s \le 24$. Eligible plans must earn at least 18 in every state. Identify eligibility throughout the range; solve for the smallest s making Maximin strictly prefer A_2 to A_3, including the boundary tie. At that grant, determine how all four recommendations change.

2. Community laundry equipment. A community laundry selects an equipment package for the coming year. Listed net profits already include purchase costs.

Payoffs

Alt.	High use	Normal use	Low use
p	0.4	0.4	0.2
A_1	120	54	6
A_2	84	66	30
A_3	54	48	42

Summary

Alt.	Max	Min	Regret	EMV
A_1	120	6	36	70.8
A_2	84	30	36	66
A_3	54	42	66	49.2

Decisions

Criterion	A_i	Value
Maximax		
Maximin		
Regret		
EMV		

A_1: Robotic package A_2: Efficient package A_3: Basic package

Task 1. Identify the four original recommendations and deciding values; include ties or mark EMV/Bayes unavailable.
Task 2. A maintenance fee changes A_1's payoffs to $(120 - f, 54 - f, 6 - f)$, where f is an integer, $0 \le f \le 6$. Financing requires EMV ≥ 64. Identify eligibility throughout the range; solve for the smallest f making Bayes strictly prefer A_2 to A_1, including the exact boundary tie. At that fee, determine how all four recommendations change.

3. Mountain excursion business. An excursion company chooses its seasonal booking capacity. A tour association may guarantee a minimum profit for its largest package.

Payoffs

Alt.	Busy season	Normal season	Quiet season
p	0.3	0.5	0.2
A_1	110	38	-18
A_2	82	62	14
A_3	58	50	32
A_4	32	32	32

Summary

Alt.	Max	Min	Regret	EMV
A_1	110	-18	50	48.4
A_2	82	14	28	58.4
A_3	58	32	52	48.8
A_4	32	32	78	32

Decisions

Criterion	A_i	Value
Maximax	_____	_____
Maximin	_____	_____
Regret	_____	_____
EMV	_____	_____

A_1: Large package A_2: Medium package A_3: Small package A_4: Fixed contract

Task 1. Identify the four original recommendations and deciding values; include ties or mark EMV/Bayes unavailable.
Task 2. A guarantee replaces A_1's quiet-season payoff -18 with an integer g, $-18 \le g \le 32$; all plans remain eligible. Derive the maximum-regret expressions and solve for the smallest g making A_1 the unique Minimax Regret choice, including the boundary tie. Find the payoff increase and determine how all four recommendations change at that g.

4. Coffee roaster contracts. A coffee roaster chooses a wholesale contract. The initial probability of strong orders is 0.30; otherwise orders are weak.

Payoffs

Alt.	Strong orders	Weak orders
p	0.3	0.7
A_1	92	8
A_2	56	38
A_3	44	42

Summary

Alt.	Max	Min	Regret	EMV
A_1	92	8	34	33.2
A_2	56	38	36	43.4
A_3	44	42	48	42.6

Decisions

Criterion	A_i	Value
Maximax	_____	_____
Maximin	_____	_____
Regret	_____	_____
EMV	_____	_____

A_1: Export contract A_2: Regional contract A_3: Local contract

Task 1. Identify the four original recommendations and deciding values; include ties or mark EMV/Bayes unavailable.
Task 2. A forecast changes the probabilities to $(p, 1-p)$, $0.30 \le p \le 0.80$; payoffs are unchanged and all contracts remain eligible. Solve the pairwise EMV inequalities for Bayes to uniquely recommend A_1, checking A_3. Give the Bayes choices below, at, and above the threshold. At $p = 0.50$, determine how all four recommendations change.

Solutions

1. Cold-storage cooperative

Task 1. Maximax: A_1 (row maximum 96); Maximin: A_3 (row minimum 34); Minimax Regret: A_2 (maximum regret 24); EMV/Bayes: A_2 (EMV 56.4).

Task 2. A_1 fails because $-12 < 18$; A_2, A_3 remain eligible for every allowed s. Since $18 + s \leq 42 < 60$, the relevant minima are $18 + s$ and 34. Thus $18 + s > 34 \iff s > 16$, so the smallest integer is $s = 17$ (\$17,000). At $s = 16$, Maximin ties at 34. At $s = 17$, eligible state bests are $(72, 60, 35)$; regret rows are $(0, 0, 0)$ and $(24, 18, 1)$. Maximax changes through eligibility and Maximin changes through the grant.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_2	72	35	0	59.8
A_3	48	34	24	42.8

Maximax: A_2 (row maximum 72); Maximin: A_2 (row minimum 35); Minimax Regret: A_2 (maximum regret 0); EMV/Bayes: A_2 (EMV 59.8).

All recommendations at the threshold boundary. Maximax: A_2 (row maximum 72); Maximin: A_2, A_3 (row minimum 34); Minimax Regret: A_2 (maximum regret 0); EMV/Bayes: A_2 (EMV 59.6).

2. Community laundry equipment

Task 1. Maximax: A_1 (row maximum 120); Maximin: A_3 (row minimum 42); Minimax Regret: A_1, A_2 (maximum regret 36); EMV/Bayes: A_1 (EMV 70.8).

Task 2. The original EMVs are 70.8, 66, and 49.2. Since the probabilities sum to one, $EMV(A_1) = 70.8 - f$. Eligibility requires $f \leq 6.8$, so A_1, A_2 qualify throughout $0 \leq f \leq 6$; A_3 never qualifies. Bayes changes when $66 > 70.8 - f \iff f > 4.8$. The boundary is a tie at $f = 4.8$; the smallest allowed integer is $f = 5$ (\$5,000). State bests at 5 are $(115, 66, 30)$, giving regret rows $(0, 17, 29)$ and $(31, 0, 0)$. Bayes changes to A_2; Maximin changes because A_3 is excluded.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	115	1	29	65.8
A_2	84	30	31	66

Maximax: A_1 (row maximum 115); Maximin: A_2 (row minimum 30); Minimax Regret: A_1 (maximum regret 29); EMV/Bayes: A_2 (EMV 66).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 115.2); Maximin: A_2 (row minimum 30); Minimax Regret: A_1 (maximum regret 28.8); EMV/Bayes: A_1, A_2 (EMV 66).

3. Mountain excursion business

Task 1. Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_2 (maximum regret 28); EMV/Bayes: A_2 (EMV 58.4).

Task 2. State bests remain $(110, 62, 32)$ throughout the range. Maximum regrets are $R_1 = \max\{24, 32 - g\}$, $R_2 = 28$, $R_3 = 52$, and $R_4 = 78$. A unique choice of A_1 requires $\max\{24, 32 - g\} < 28 \iff g > 4$. At $g = 4$, A_1, A_2 tie with maximum regret 28. The smallest integer is $g = 5$; the increase is $5 - (-18) = 23$ (\$23,000). At 5, A_1 has regrets $(0, 24, 27)$ and EMV $33 + 19 + 1 = 53$. Minimax Regret changes to A_1; Maximin retains its two-way tie.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	110	5	27	53
A_2	82	14	28	58.4
A_3	58	32	52	48.8
A_4	32	32	78	32

Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_1 (maximum regret 27); EMV/Bayes: A_2 (EMV 58.4).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_1, A_2 (maximum regret 28); EMV/Bayes: A_2 (EMV 58.4).

4. Coffee roaster contracts

Task 1. Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_2 (EMV 43.4).

Task 2. $E_1 = 8 + 84p$, $E_2 = 38 + 18p$, $E_3 = 42 + 2p$. Hence $E_1 > E_2 \iff p > 5/11$ and $E_1 > E_3 \iff p > 17/41$. Also $E_2 > E_3 \iff p > 1/4$, true throughout the allowed range. Bayes selects A_2 for $0.30 \le p < 5/11$, ties A_1, A_2 at $p = 5/11$ (EMV 508/11), and selects A_1 for $5/11 < p \le 0.80$. At the boundary $E_3 = 472/11 < 508/11$. At $p = 0.50$, EMVs are 50, 47, 43. Only Bayes changes: payoffs, extrema, and regrets are unchanged.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	92	8	34	50
A_2	56	38	36	47
A_3	44	42	48	43

Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_1 (EMV 50).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_1, A_2 (EMV $\frac{508}{11}$).